

## Chapter 3 Describing, Exploring, and Comparing Data

### Measures of Relative Standing and Boxplots (3.3)

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## Objectives

In this lesson, we will learn how to

- compute and interpret **z scores**;
- use **z scores** to compare values from different data sets;
- identify significantly low and significantly high values;
- find and interpret percentiles and quartiles;
- find a five-number summary;
- construct and interpret boxplots;
- identify outliers and construct modified boxplots.

## 1. Measures of Relative Standing

A **measure of relative standing** describes the location of a data value relative to the other values in the same data set.

The most important measure of relative standing in this section is the **z score**.

### **z Score**

A **z score**, also called a **standard score** or **standardized value**, is the number of standard deviations that a value  $x$  is above or below the mean.

For sample data,

$$z = \frac{x - \bar{x}}{s}$$

For population data,

$$z = \frac{x - \mu}{\sigma}$$

### Round-Off Rule for z Scores

Round z scores to

two decimal places,

such as

$$z = 2.31.$$

## 2. Important Properties of z Scores

### Interpreting z Scores

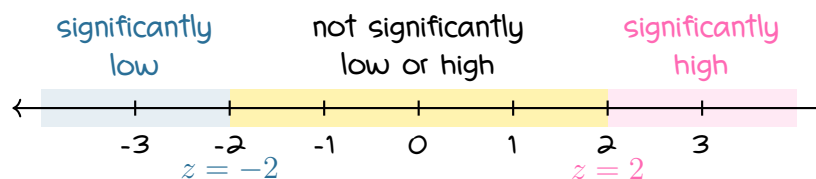
- A positive z score means the value is above the mean.
- A negative z score means the value is below the mean.
- A z score of 0 means the value is equal to the mean.
- z scores have no units.

### Significant Values Using z Scores

$$z \leq -2 \implies \text{significantly low}$$

$$z \geq 2 \implies \text{significantly high}$$

$$-2 < z < 2 \implies \text{not significantly low or high}$$



### Comparing Values from Different Data Sets

The absolute value

$$|z|$$

measures how far a value is from its mean in standard-deviation units. Therefore,

$$\text{larger } |z| \implies \text{more extreme value}$$

Example 1:

Which of the following two data values is more extreme relative to the data set from which it came?

- A 99°F body temperature among 106 adults with

$$\bar{x} = 98.20^\circ\text{F}, \quad s = 0.62^\circ\text{F}.$$

- A 5.7790 g quarter weight among 40 quarters with

$$\bar{x} = 5.63930 \text{ g}, \quad s = 0.06194 \text{ g}.$$

Solution

For the body temperature,

$$z = \frac{x - \bar{x}}{s} = \frac{99 - 98.20}{0.62} = \frac{0.80}{0.62} \approx 1.29.$$

For the quarter weight,

$$\begin{aligned} z &= \frac{x - \bar{x}}{s} = \frac{5.7790 - 5.63930}{0.06194} \\ &= \frac{0.13970}{0.06194} \approx 2.26. \end{aligned}$$

Compare the absolute values:

$$|1.29| = 1.29$$

and

$$|2.26| = 2.26.$$

Because

$$2.26 > 1.29,$$

the quarter weight is farther from its mean in standard-deviation units.  
Therefore,

the quarter weight is more extreme.

Example 2:

Among 600 earthquake magnitudes, the mean is

$$\bar{x} = 2.572$$

and the standard deviation is

$$s = 0.651.$$

One earthquake has magnitude

$$4.01.$$

Is the magnitude 4.01 significantly high?

Solution

Compute the z score:

$$\begin{aligned} z &= \frac{x - \bar{x}}{s} = \frac{4.01 - 2.572}{0.651} \\ &= \frac{1.438}{0.651} \approx 2.21. \end{aligned}$$

Since

$$2.21 \geq 2,$$

the value is significantly high.

Therefore,

magnitude 4.01 is significantly high.

### 3. Percentiles

Percentiles

Percentiles are measures of location, denoted

$$P_1, P_2, \dots, P_{99},$$

which divide a data set into 100 groups with about 1% of the values in each group.

Finding the Percentile of a Data Value

For a data value  $x$ ,

$$\text{Percentile of } x = \frac{\text{number of values less than } x}{\text{total number of values}} \cdot 100$$

Round the result to the nearest whole number.

Example 3:

The following are 50 sorted Space Mountain wait times, in minutes.

|    |    |    |    |    |    |    |    |     |     |
|----|----|----|----|----|----|----|----|-----|-----|
| 10 | 15 | 15 | 15 | 15 | 15 | 20 | 20 | 20  | 20  |
| 25 | 25 | 25 | 25 | 25 | 25 | 30 | 30 | 30  | 30  |
| 30 | 30 | 30 | 30 | 35 | 35 | 35 | 35 | 35  | 35  |
| 35 | 35 | 40 | 40 | 40 | 40 | 45 | 50 | 50  | 50  |
| 50 | 50 | 55 | 55 | 60 | 75 | 75 | 75 | 105 | 110 |

Find the percentile for a wait time of 45 minutes.

Solution

There are

36

wait times less than 45 minutes, and there are

50

total wait times.

Therefore,

$$\text{Percentile of } 45 = \frac{36}{50}(100) = 72.$$

So 45 minutes is in the

72nd percentile.

Loosely speaking, a wait time of 45 minutes separates the lowest 72% of values from the highest 28%.

Thus,

$$P_{72} = 45 \text{ minutes.}$$

## 4. Converting a Percentile to a Data Value

### Notation

- $n$ : total number of values in the data set;
- $k$ : percentile being used;
- $L$ : locator giving the position of a value;
- $P_k$ :  $k$ th percentile.

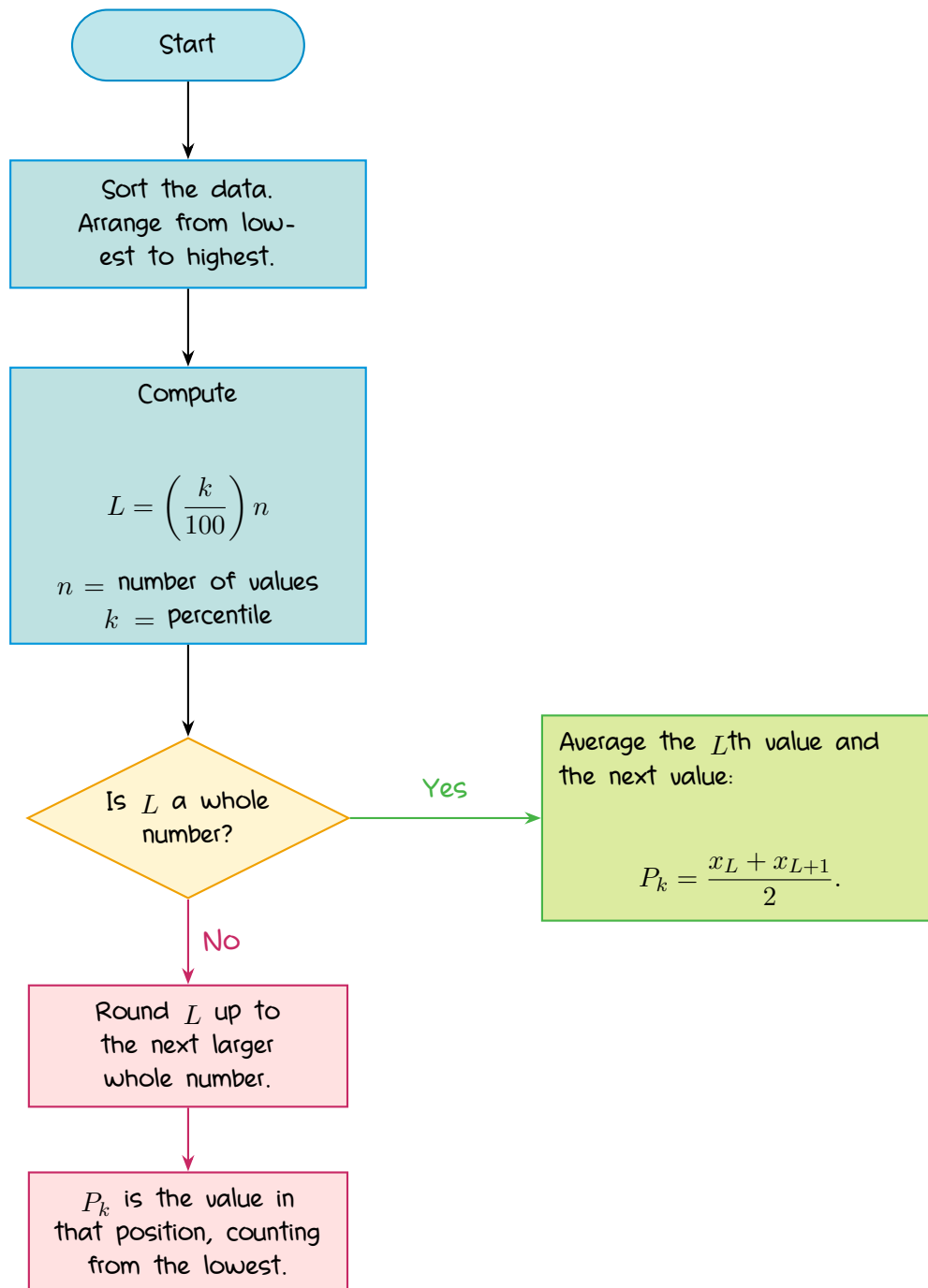
### Locator Formula

To find the data value corresponding to the  $k$ th percentile, compute

$$L = \frac{k}{100}n$$

Then use one of two cases:

- If  $L$  is not a whole number, round  $L$  up to the next larger whole number.  $P_k$  is the value in that position.
- If  $L$  is a whole number,  $P_k$  is the mean of the values in positions  $L$  and  $L + 1$ .



### Two Cases to Remember

|               |                                       |
|---------------|---------------------------------------|
| $L$ not whole | → round up and use that position,     |
| $L$ whole     | → average positions $L$ and $L + 1$ . |

Example 4:

Use the 50 sorted Space Mountain wait times to find

$$P_{25}.$$

### Solution

Here,

$$k = 25 \quad \text{and} \quad n = 50.$$

Compute the locator:

$$L = \frac{25}{100}(50) = 12.5.$$

Since  $L = 12.5$  is not a whole number, round up to

$$13.$$

The 13th value in the sorted data is

$$25.$$

Therefore,

|                                |
|--------------------------------|
| $P_{25} = 25 \text{ minutes}.$ |
|--------------------------------|

## 5. Quartiles

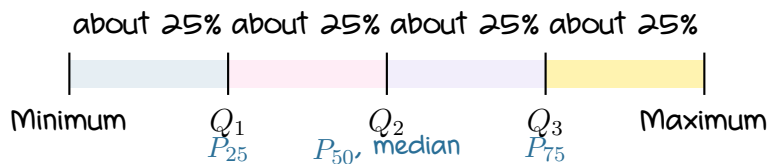
### Quartiles

Quartiles divide a sorted data set into four groups with about 25% of the values in each group.

$$Q_1 = P_{25}$$

$$Q_2 = P_{50} = \text{median}$$

$$Q_3 = P_{75}$$



### Interpreting Quartiles

- $Q_1$  separates approximately the lowest 25% from the highest 75%.
- $Q_2$  separates approximately the lowest 50% from the highest 50%.
- $Q_3$  separates approximately the lowest 75% from the highest 25%.

### Important: Quartile Methods Can Differ

There is no universal agreement on one procedure for finding percentiles and quartiles.

Different calculators and software packages may therefore produce slightly different values for  $Q_1$  and  $Q_3$ .

For this course, use the procedure presented in these notes unless

otherwise instructed.

### Statistics Using Quartiles and Percentiles

$$\text{Interquartile Range} = IQR = Q_3 - Q_1$$

$$\text{Semi-interquartile Range} = \frac{Q_3 - Q_1}{2}$$

$$\text{Midquartile} = \frac{Q_1 + Q_3}{2}$$

$$10\text{--}90 \text{ Percentile Range} = P_{90} - P_{10}$$

## 6. Five-Number Summary

### Five-Number Summary

For a data set, the five-number summary consists of:

1. Minimum
2. First quartile,  $Q_1$
3. Second quartile,  $Q_2$ , the median
4. Third quartile,  $Q_3$
5. Maximum

Minimum,  $Q_1$ ,  $Q_2$ ,  $Q_3$ , Maximum

Example 5:

Use the 50 sorted Space Mountain wait times to find the five-number summary.

Solution

From the sorted data,

$$\text{Minimum} = 10$$

and

$$\text{Maximum} = 110.$$

Using the percentile procedure,

$$Q_1 = 25, \quad Q_2 = 35, \quad Q_3 = 50.$$

Therefore, the five-number summary is

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 10, | 25, | 35, | 50, | 110 |
|-----|-----|-----|-----|-----|

minutes.

## 7. Boxplots

Boxplot

A boxplot, or box-and-whisker diagram, is a graph based on the five-number summary.

It contains:

- a line extending from the minimum to the maximum;
- a box extending from  $Q_1$  to  $Q_3$ ;
- a line inside the box at the median  $Q_2$ .

### Procedure for Constructing a Boxplot

1. Find the five-number summary.
2. Draw a number line covering the range of the data.
3. Draw a box from  $Q_1$  to  $Q_3$ .
4. Draw a vertical line inside the box at  $Q_2$ .
5. Extend the graph to the minimum and maximum values.

Example 6:

Construct a boxplot for the Space Mountain wait times using

10, 25, 35, 50, 110.

### Solution

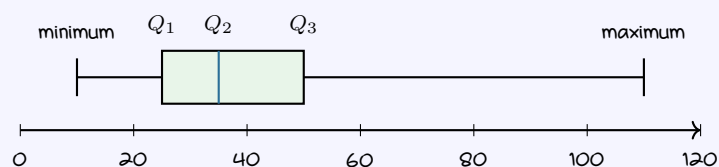
The five-number summary is

Minimum = 10,

$Q_1 = 25$ ,       $Q_2 = 35$ ,       $Q_3 = 50$ ,

and

Maximum = 110.



### Reading a Boxplot

A boxplot is not just a picture of five numbers.

It also divides the data into four sections.

#### About 25% in Each Section

Approximately

25%

of the data lie between the minimum and  $Q_1$ ,  
approximately

25%

lie between  $Q_1$  and the median,  
approximately

25%

lie between the median and  $Q_3$ ,  
and approximately

25%

lie between  $Q_3$  and the maximum.

#### The Box Contains the Middle 50%

Because the box extends from  $Q_1$  to  $Q_3$ ,

about 50% of the data lie inside the box

and its length is the

$$IQR = Q_3 - Q_1.$$

#### Interpretation Question

For the Space Mountain data,

$$Q_1 = 25 \quad \text{and} \quad Q_3 = 50.$$

Approximately what percentage of wait times lie between 25 and 50 minutes?

Solution

The interval from  $Q_1$  to  $Q_3$  contains the middle half of the observations. Therefore,

about 50%

of the wait times lie between 25 and 50 minutes.

## 8. Shape and Boxplots

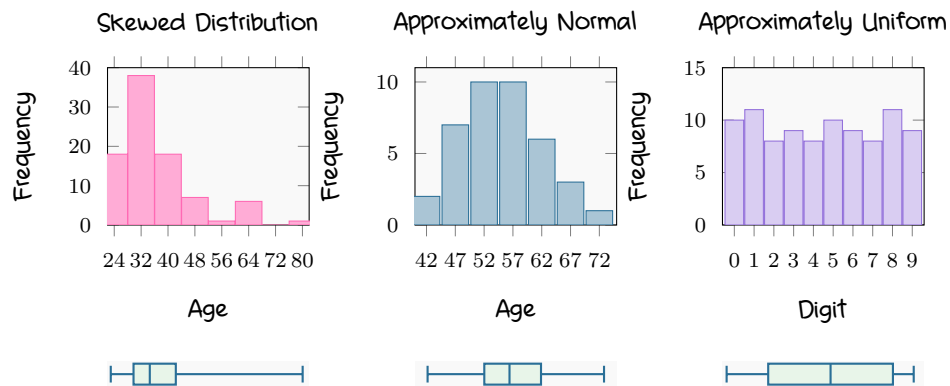
A boxplot can often help us recognize whether a distribution is symmetric or skewed.

### Reading Shape from a Boxplot

Look at the whole boxplot:

- the relative lengths of the whiskers;
- the position of the median inside the box;
- the relative lengths of the two halves of the box.

A noticeably longer side often points in the direction of the longer tail of the distribution.



## 9. Modified Boxplots and Outliers

A regular boxplot uses the minimum and maximum.

A modified boxplot treats outliers differently.

### Identifying Outliers

First compute

$$IQR = Q_3 - Q_1.$$

Then calculate

$$1.5(IQR).$$

The two cutoff values, called fences, are

$$\text{Lower Fence} = Q_1 - 1.5(IQR)$$

and

$$\text{Upper Fence} = Q_3 + 1.5(IQR).$$

A value below the lower fence or above the upper fence is classified as an outlier.

### The Fences Are Not Whisker Endpoints

The lower and upper fences are cutoff values used to identify outliers. They do not have to be actual observations. The whiskers extend to the smallest and largest observed values that are not outliers.

### Modified Boxplot

A modified boxplot differs from a regular boxplot in two ways:

1. Outliers are shown with separate symbols, such as dots.
2. The whiskers extend only to the smallest and largest values that are not outliers.

Example 7:

For the Space Mountain wait times,

$$Q_1 = 25, \quad Q_2 = 35, \quad Q_3 = 50.$$

Determine the outliers and construct a modified boxplot.

### Solution

First find the interquartile range:

$$IQR = Q_3 - Q_1 = 50 - 25 = 25.$$

Now compute

$$1.5(IQR) = 1.5(25) = 37.5.$$

The lower fence is

$$Q_1 - 1.5(IQR) = 25 - 37.5 = -12.5.$$

The upper fence is

$$Q_3 + 1.5(IQR) = 50 + 37.5 = 87.5.$$

Therefore, a value is an outlier if it is

$$x < -12.5$$

or

$$x > 87.5.$$

The two largest observations are

$$105 \quad \text{and} \quad 110.$$

Because

$$105 > 87.5$$

and

$$110 > 87.5,$$

both are outliers.

There are no observations below  $-12.5$ .

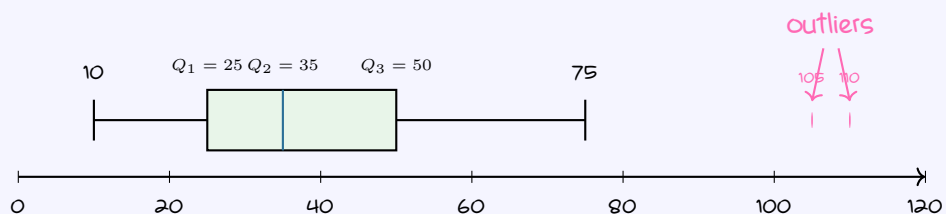
The smallest non-outlier is

$$10,$$

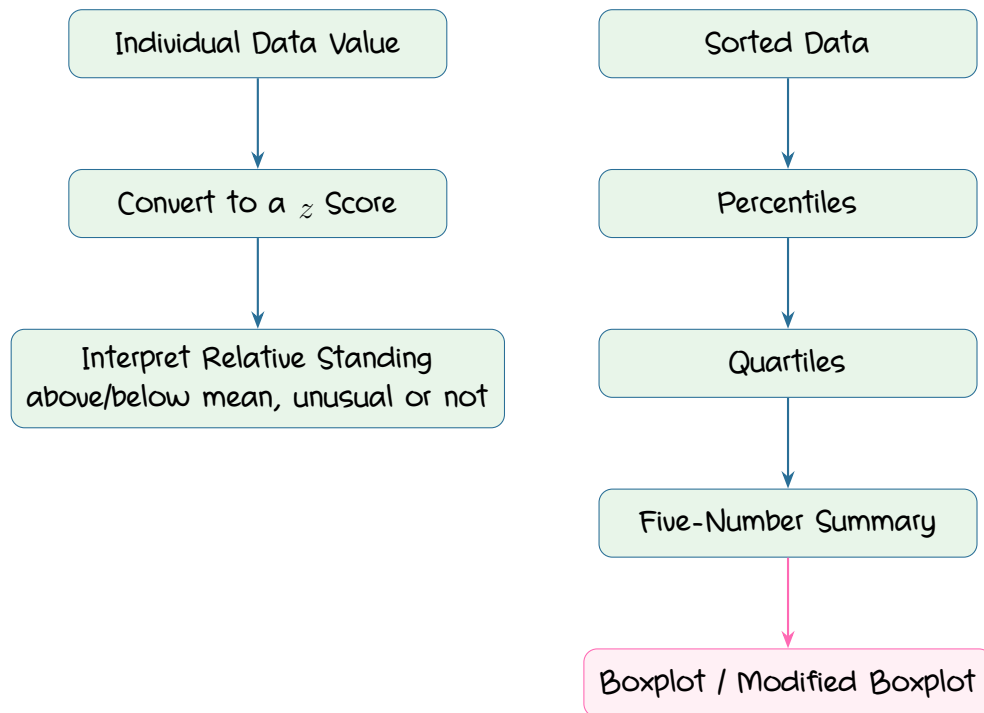
and the largest non-outlier is

$$75.$$

Thus the modified boxplot has whiskers at 10 and 75, with separate points at 105 and 110.



## 10. The Big Picture



## Summary

### Key Ideas

- A **z score** tells how many standard deviations a value is above or below the mean.
- Positive z scores correspond to values above the mean, and negative z scores correspond to values below the mean.
- To compare values from different data sets, compare  $|z|$ .
- Values with

$$z \leq -2$$

are significantly low, and values with

$$z \geq 2$$

are significantly high.

- Percentiles divide sorted data into 100 groups with about 1% in each group.
- Quartiles divide sorted data into four groups with about 25% in each group.

- 

$$Q_1 = P_{25}, \quad Q_2 = P_{50}, \quad Q_3 = P_{75}.$$

- The five-number summary is

minimum,  $Q_1$ , median,  $Q_3$ , maximum.

- A boxplot displays the five-number summary visually.
- The box from  $Q_1$  to  $Q_3$  contains approximately the middle 50% of the observations.

- 

$$IQR = Q_3 - Q_1.$$

- Outliers are identified using the fences

$$Q_1 - 1.5(IQR)$$

and

$$Q_3 + 1.5(IQR).$$

- A modified boxplot displays outliers separately and extends its whiskers only to non-outlier values.